MAT2002 **Name (Print):** _____
Spring 2026
HW6
Deadline: 7th May

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1. The Gamma Function. The gamma function is denoted by $\Gamma(p)$ and is defined by the integral

$$\Gamma(p+1) = \int_0^\infty e^{-x} x^p dx \tag{i}$$

The integral converges as $x \to \infty$ for all p. For $p < 0$ it is also improper at $x = 0$, because the integrand becomes unbounded as $x \to 0$. However, the integral can be shown to converge at $x = 0$ for $p > -1$.

(a) Show that, for $p > 0$,

$$\Gamma(p+1) = p\Gamma(p).$$

(b) Show that $\Gamma(1) = 1$.

(c) If p is a positive integer n, show that

$$\Gamma(n+1) = n!.$$

Since $\Gamma(p)$ is also defined when p is not an integer, this function provides an extension of the factorial function to nonintegral values of the independent variable. Note that it is also consistent to define $0! = 1$.

(d) Show that, for $p > 0$,

$$p(p+1)(p+2)\cdots(p+n-1) = \Gamma(p+n)/\Gamma(p).$$

Thus $\Gamma(p)$ can be determined for all positive values of p if $\Gamma(p)$ is known in a single interval of unit length-say, $0 < p \leq 1$. It is possible to show that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. Find $\Gamma\left(\frac{3}{2}\right)$ and $\Gamma\left(\frac{11}{2}\right)$.

2. Consider the Laplace transform of t^p, where $p > -1$.

(a) Referring to Problem 1, show that

$$\begin{aligned}\mathcal{L}\{t^p\} &= \int_0^\infty e^{-st} t^p dt = \frac{1}{s^{p+1}} \int_0^\infty e^{-x} x^p dx \\ &= \Gamma(p+1)/s^{p+1}, \quad s > 0\end{aligned}$$

(b) Let p be a positive integer n in part (a); show that

$$\mathcal{L}\{t^n\} = n!/s^{n+1}, \quad s > 0$$

(c) Show that

$$\mathcal{L}\left\{t^{-1/2}\right\} = \frac{2}{\sqrt{s}} \int_0^\infty e^{-x^2} dx, \quad s > 0$$

It is possible to show that

$$\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$$

hence

$$\mathcal{L}\left\{t^{-1/2}\right\} = \sqrt{\pi/s}, \quad s > 0.$$

(d) Show that

$$\mathcal{L}\left\{t^{1/2}\right\} = \sqrt{\pi}/\left(2s^{3/2}\right), \quad s > 0.$$

3. Solve the given initial value problem with Laplace transformation

(i) $y'' + 2y' + 5y = 0; \quad y(0) = 2, \quad y'(0) = -1$

(ii) $y^{(4)} - 4y''' + 6y'' - 4y' + y = 0; \quad y(0) = 0, \quad y'(0) = 1, \quad y''(0) = 0, \quad y'''(0) = 1$

4. Find the Laplace transformation for $f(t) = \begin{cases} 0, & t < \pi \\ t - \pi, & \pi \le t < 2\pi \\ 0, & t \ge 2\pi \end{cases}$

5. Find the solution of the given initial value problem:

(i) $y'' + y' + \frac{5}{4}y = g(t); \quad y(0) = 0, \quad y'(0) = 0; \quad g(t) = \begin{cases} \sin t, & 0 \le t < \pi \\ 0, & t \ge \pi \end{cases}$

(ii) $y^{(4)} - y = u_1(t) - u_2(t); \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 0, \quad y'''(0) = 0$

6. Find the solution of the given initial value problem:

(i) $y'' - y = -20\delta(t-3); \quad y(0) = 1, \quad y'(0) = 0$

(ii) $y'' + 2y' + 2y = \cos t + \delta(t - \pi/2); \quad y(0) = 0, \quad y'(0) = 0$

7. Consider the equation

$$\phi(t) + \int_0^t k(t-\xi)\phi(\xi)d\xi = f(t)$$

in which f and k are known functions, and ϕ is to be determined. Since the unknown function ϕ appears under an integral sign, the given equation is called an integral equation; in particular, it belongs to a class of integral equations known as Volterra integral equations. Take the Laplace transform of the given integral equation and obtain an expression for $\mathcal{L}\{\phi(t)\}$ in terms of the transforms $\mathcal{L}\{f(t)\}$ and $\mathcal{L}\{k(t)\}$ of the given functions f and k. The inverse transform of $\mathcal{L}\{\phi(t)\}$ is the solution of the original integral equation.

8. Consider the Volterra integral equation (see Problem 7)

$$\phi(t) + \int_0^t (t-\xi)\phi(\xi)d\xi = \sin 2t \tag{ii}$$

(a) Solve the integral equation (ii) by using the Laplace transform.

(b) By differentiating Eq. (ii) twice, show that $\phi(t)$ satisfies the differential equation

$$\phi''(t) + \phi(t) = -4\sin 2t.$$

Show also that the initial conditions are

$$\phi(0) = 0, \quad \phi'(0) = 2.$$

(c) Solve the initial value problem in part (b), and verify that the solution is the same as the one in part (a).